

Shri Guru Gobind Singhji Institute of Engineering and Technology, Vishnupuri, Nanded (Maharashtra State)- 431 606

DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION ENGINEERING

Curriculum of M. Tech. (Embedded Systems and VLSI Design)

(With effective from 2023-2024)

First Year					
Course Code	Name of the course	L	T	P	Credits
Semester I					
PCC-EV-501	Semiconductor Devices	3	0	2	4
PCC-EV-503	Real Time Embedded Systems	3	0	2	4
PCC-EV-505	Digital IC Design	3	0	2	4
PEC-EV-5AA	Programme Specific Elective Course-I (Students can register for any one course offered by the department from the list given below)	3	0	2	4
MCC-590	Research Methodology and IPR	2	0	0	2
MAC-591	English for Research Paper Writing	2	0	0	2
PRJ-EV-551	Mini Project and Seminar	0	0	4	2
Sub-Total		16	0	12	22
Semester II					
PCC-EV-502	Analog and Mixed Signal VLSI Design	3	0	0	3
PCC-EV-504	Advanced Verification Techniques using System Verilog	3	0	2	4
PEC-EV-5BB	Programme Specific Elective Course-II (Students can register for any one course offered by the department from the list given below)	3	0	2	4
PEC-EV-5CC	Programme Specific Elective Course-III (Students can register for any one course offered by the department from the list given below)	3	0	2	4
PRJ-EV-552	Mini Project and Seminar	0	0	4	2
OEC-8**	Open Elective (Students can register for any one course offered by the institute from the list given below)	3	0	0	3
AUD-9**	Audit Course (Students can register for any one course offered by the institute from the list given below)	2	0	0	--
Sub-Total		17	0	10	22
Second Year					
Semester I					
DIS-EV-601	Dissertation - I	--	--	--	18
HSSM-601	Indian Knowledge System [#]	2	--	--	2
Sub-Total		2	--	--	20
Semester II					
DIS-EV-602	Dissertation – II	--	--	--	20
CCC-602	Yoga and Sports [#]	--	--	2	--
Sub-Total		--	--	2	20
Grand Total					

L = No. of Lecture Hours/week, T = No. of Tutorial Hours/week, P = No. of Practical Hours/week

[#]These courses will be offered in online/mixed mode to facilitate internship. Students are also encouraged to take quality MOOC courses in lieu of these courses, such as NPTEL, SWAYAM, Coursera, etc.

In addition to these listed courses, student can choose over and above the open electives and courses offered in the institute at institute level.

Program Elective – I (Students can register for any one course offered by the department from the list given below)	
Course Code	Name of the course
PEC-EV-507	Modern Digital Design using Verilog
PEC-EV-509	Digital Signal and Image Processing
PEC-EV-511	Object Oriented Programming
PEC-EV-513	Computer Architecture
PEC-EV-515	High Speed Digital Design
PEC-EV-517	CAD of Digital System
PEC-EV-519	IC Technology
PEC-EV-521	Parallel Processing
PEC-EV-523	Programmable Digital Signal Processors
Program Elective – II (Students can register for any one course offered by the department from the list given below)	
Course Code	Name of the course
PEC-EV-506	Low Power VLSI Design
PEC-EV-508	Memory Technologies
PEC-EV-510	Physical Design Automation
PEC-EV-512	Testability of VLSI Circuits
PEC-EV-514	VLSI Signal Processing
PEC-EV-516	SoC Design
Program Elective – III (Students can register for any one course offered by the department from the list given below)	
Course Code	Name of the course
PEC-EV-518	Embedded Networking
PEC-EV-520	System Design with Embedded OS
PEC-EV-522	Communication Buses and Interfaces
PEC-EV-524	Network Security and Cryptography
Open Elective Courses (Students can register for any one course offered by the institute from the list given below)	
Course Code	Name of the course
OEC-801	Business Analytics
OEC-802	Industrial Safety
OEC-803	Operations Research
OEC-804	Cost Management of Engineering Projects
OEC-805	Composite Materials
OEC-806	Waste to Energy
OEC-807	Cyber Security
Audit Courses (Students can register for any one course offered by the institute from the list given below)	
Course Code	Name of the course
AUD-901	Project Management
AUD-902	Disaster Management
AUD-903	Sanskrit for Technical Knowledge
AUD-904	Value Education
AUD-905	Constitution of India
AUD-906	Pedagogy Studies
AUD-907	Stress Management by Yoga
AUD-908	Personality Development through Life Enlightenment Skills

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